

When is one field enough?

A measurement audit of an organ-on-a-chip quality-control benchmark, a corrected evaluation protocol, and a chip-level QC tool with an explicit "inconclusive" outcome.

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Code, video, demo	https://github.com/Foreist/when-is-one-field-enough · corrected demo v10 https://taewoong23-ooc-chip-qc-demo.static.hf.space/video.html · direct MP4 https://taewoong23-ooc-chip-qc-demo.static.hf.space/demo_video.mp4 · browser demo in \$6.8

Summary

Organ-on-a-chip (OoC) cultures are inspected daily under a microscope, and the quality-control (QC) decision — keep the chip, re-image, or discard it — is still made by eye. A public benchmark exists for automating this: the **OOC Image Dataset** (3,072 brightfield fields, 59 acquisition sessions, six cell lines, labels from four expert cell biologists by majority vote). Before building on it we asked a simple question: *is this benchmark measuring what it claims to measure?*

It is not. Four measured findings:

- 1. The published split leaks sessions.** Training and test images come from the *same 57 of 59 sessions*. Holding the test images and the training-set size fixed, and changing only whether the test sessions also contribute training images, a **lighter** 224 px / 6-epoch recipe on 20 test sessions moves accuracy by **+7.9 pp (95% CI 4.2–11.5)** and AUC by **+8.9 pp (95% CI 6.7–11.0)** across eight seeds — not a re-score of the published split's own headline. The shipped split reports 78.9%; the same model on unseen sessions reports 56.0%, below always predicting the commoner label there (65.7%).
- 2. The 3,072 labels are not 3,072 independent observations.** Labels are strongly autocorrelated along the acquisition order (session ICC 0.321; in mixed sessions runs average 5.42 fields versus 2.50 with the labels shuffled). The design effect is 17–38 depending on how sessions of unequal size are weighted, so **label prevalence** has effective N roughly **80–180** (not an accuracy/AUC N) — and a single field agrees with the session's majority label only **78.8%** of the time (91.0% even with ten fields).
- 3. The failure mode is contiguous in acquisition order, not per-image — and our own policy comparison did not survive its own criticism.** Failures occupy contiguous stretches of the filename-index order (runs test significant in 23/45 mixed sessions, surviving a cell-type control); that is not a mapped XY spatial block. A policy comparison simulated on the *labels* suggests random fields are best for the chip call; **with the model in the loop that ranking disappears** — over 13–15 chips no paired difference survives a correction for the six comparisons made (§4.3). What does survive is a concrete, reproducible failure: reading the **first *k*** fields called a 100%-bad chip *pass* with 0.94 confidence, because the start of a session can sit inside a good stretch of the order. The shipped tool reads an evenly spaced grid of fields for that reason, not on the strength of a policy ranking.
- 4. The leakage is not unique to this benchmark.** Auditing a second public organoid imaging benchmark (OCT organoid tracking, zenodo.15783866) shows the same defect: **40.0% of its test images belong to a (well, day) acquisition group that also appears in training** (§4.4).

We then built the tool the corrected protocol implies. It accepts a folder of fields as one chip; measured numbers are on **session-derived half-chip proxies** (random withheld halves of dates — a date is one microscope run over one or more chips), not verified physical chips. On 25 such proxies it reaches the same chip-level accuracy as reading **every** field (0.800) while using **9.5 fields instead of 27.4** — 2.9× fewer **decision** fields on scored maps, not microscope time — and defers 8% of chips to a human rather than guessing. Fields are read from an **evenly spaced grid** (every field in turn on chips of ≤ 20 fields; reading the first k of a long chip can sit inside a good stretch of the acquisition order and stop early — we observed a 100%-bad chip called *pass* with 0.94 confidence), and a sequential stopping rule returns **pass / fail / inconclusive** with a posterior confidence. On 25 unseen proxies: **82.6% chip-level accuracy on the chips it calls (23 of 25), 13% of calls confident-but-wrong, 8% inconclusive, 9.5 fields on average**. A plain cap of 12 spread fields per chip does as well on field count (9.0 on average, 0.800); the saving comes from not reading every field, and what the sequential rule adds is a posterior confidence and the option to defer. The 8-field minimum behind these numbers was tuned on the same 25 chips; re-selected by cross-validation on the other 34 sessions it would be 1, which scores 0.680 here with every chip called (0.708 on the 24 it calls; §6.2(c)). On those 68 OOF half-session proxies (per-fold retraining; not 68 new physical chips) the rule still matches reading every field with a fifth to two-fifths of the fields.

We also report two things we could not do, because they define the honest boundary of this work: the **re-image vs discard** distinction is *not* supported by the data (a constructed recovery target is not predictable; CV AUC 0.613 against a permutation-null 95th percentile of 0.564), and **no reliable out-of-distribution detector** was found (image statistics and feature-space distance both missed the worst failure). Finally, simulating a stopping rule on ground-truth labels **overstates** it: on the same 25 chips, without a minimum-field guard, labels give 0.875 and the model 0.708.

Everything is reproducible from one public dataset with the scripts in this repository.

1. Introduction

1.1 The problem

Microphysiological systems — organ-on-a-chip, organoids in microfluidic devices — are promoted as human-relevant alternatives to animal testing. One recurring practical step is quality control: a chip is imaged repeatedly over days, and at each session someone must decide whether the fields in front of them are usable. Technical artifacts (air bubbles, defocus, deformed channel walls) and biological problems (cells not attaching, density far from the expected range) both appear as "bad", and both are judged by eye, field by field.

Automating this judgement is attractive, and a public benchmark for it exists. But an automated QC system inherits every property of its benchmark: if the benchmark's split leaks, reported accuracy is inflated; if its labels are clustered, its effective sample size is far smaller than it looks; if its labels are field-dependent, then "accuracy per image" is not the quantity a lab actually needs. The benchmark we audit reports none of these checks.

Who this is for. (i) *Chip and organoid labs* that image cultures daily and decide keep / re-image / discard by eye; (ii) *imaging and data teams* building reusable, trustworthy culture datasets; (iii) *assays downstream of QC* — dose-response, toxicity, drug evaluation — which inherit the quality of the fields that were kept.

1.2 What this report does

We take one public OoC QC benchmark — the OOC Image Dataset [1,2] — and treat the benchmark itself as the object of study. We (i) audit its split and label structure, (ii) derive and validate a corrected evaluation protocol, (iii) measure how sampling and aggregation policies behave on the measured structure, and (iv) build a chip-level QC tool and evaluate it on held-out sessions, reporting which of its settings were chosen on the test chips (§6.2(c)).

1.3 What we claim — and what we do not

We claim **measurement results** about a specific benchmark and a **protocol/tool** that follows from them. We do *not* claim a new model architecture, biological or clinical validity, or generalisation beyond this dataset. Leakage and label clustering are known phenomena in machine learning [7–11]; our contribution is not their discovery but their **measured magnitude in this benchmark, the corrected protocol, and the field-sampling policy we adopted after one observed failure** — together with two negative results that bound the claims. The policy is not a measured optimum: a ranking of sampling policies did not survive the model in the loop (§4.3).

2. Data

Source. OOC Image Dataset, Zenodo record 10.5281/zenodo.10203721 [3]; data descriptor in *Data* 2024 [1] and a companion conference paper [2]. The Zenodo record states **CC-BY-4.0**; the MDPI descriptor lists **CC-BY-SA**. We resolve the ambiguity conservatively: we redistribute no images beyond the 44 attributed demo example fields, license our code MIT and our derived artefacts CC-BY-SA.

Content. 3,072 brightfield images ($2,056 \times 1,542$ px) from an automated microscope on an OoC setup, covering six cell lines (A549, Caco-2, HPMEC, HUVEC, NHBE, HSAEC). Filenames are `YYMMDD_N.png`: YYMMDD identifies an **acquisition session** (59 in total) and N the field number within it. Each image carries a binary label, **good** or **bad**, assigned by **four expert cell biologists**, with disagreement resolved by majority vote or an additional rater [1]. Metadata (cell type, seeding density, flow rate, day after seeding) is available for part of the data. Figure 1 summarises the corpus: sessions range from 6 to 229 fields (median 24), and the share of fields labelled **bad** ranges from 0% to 100% per session.

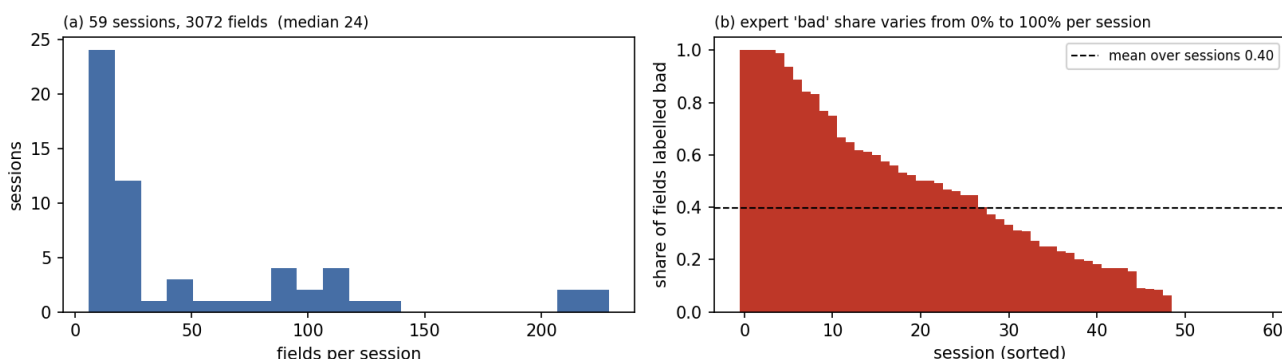


Figure 1. The corpus. (a) fields per session; (b) per-session share of fields labelled **bad**.

The published split. The zip ships train/val/test folders (2,130 / 286 / 656 images). This is the split used by the reference baseline in [1].

Privacy, ethics and compliance. The dataset contains brightfield images of immortalised cell lines grown in microfluidic devices. There are no human subjects, no patient-derived material, no identifiable personal data and no clinical records, so no consent or IRB review applies; the data are distributed publicly under the licences above. The tool automates a quality judgement about an image — it makes no biological, diagnostic or clinical claim, and we do not use it to make decisions about patient material. The second benchmark audited in §4.4 is likewise public (CC-BY-4.0) and contains only instrument images.

3. Methods

All quantities below are defined once here and used unchanged throughout; the scripts that compute them are listed in §10.

Sessions and fields. A *session* is one acquisition date (YYMMDD in the filename) — one microscope run over one or more chips. A *field* is one image. Field index N is a proxy for acquisition order, not XY; a *run* is a maximal contiguous equal-label block in that order. The tool accepts a folder of fields as one chip; chip-level metrics below are on **session-derived half-chip proxies** (random withheld halves of dates), not verified physical chips.

Runs test (clustering). For a binary sequence with n_1 bad and n_0 good fields, the number of runs R has, under independence, mean $\mu = 2n_1n_0/n + 1$ and variance $\sigma^2 = 2n_1n_0(2n_1n_0 - n) / (n^2(n - 1))$; we report $z = (R - \mu)/\sigma$ and the two-sided normal p-value. Negative z means fewer runs than chance, i.e. clustering. Sessions without both labels are skipped. The i.i.d. expected run length is $1 / (2p(1-p))$ with p the observed good fraction.

Intraclass correlation and design effect. For clusters (sessions) with sizes n_i , ICC(1) is estimated by one-way ANOVA, $ICC = (MSB - MSW) / (MSB + (m_0 - 1) \cdot MSW)$ with $m_0 = (N - \sum n_i^2/N)/(k - 1)$; the design effect is $1 + (m_0 - 1) \cdot ICC$ and the effective sample size is $N / \text{design effect}$; Kish's $\bar{m} = \sum n_i^2/N$ is used for the **label-prevalence** mean, not for accuracy or AUC.

Controlled leakage A/B. Test images are fixed by choosing 20 sessions and withholding half of each session's fields as the test set. The other half of those sessions' fields are *available* but used only in the leaky arm. Both arms receive exactly N training fields (N = the non-test pool minus the 200 shared validation fields): the disjoint arm draws all N from non-test sessions; the leaky arm draws $N - k$ from non-test sessions and k from the available test-session fields. Model, budget, augmentation and seeds are identical. This is a lighter 224 px / 6-epoch recipe, not a re-training of the published split.

Chip-level decision. For a set of fields with per-field bad probabilities p_j , the sequential rule maintains a $\text{Beta}(1,1)$ posterior on the bad fraction f — $f \mid \text{data} \sim \text{Beta}(1 + b, 1 + g)$ with b, g the counts of fields with $p_j > 0.5$ and ≤ 0.5 — and stops as soon as $P(f > 0.5) > 0.9$ or $P(f < 0.5) > 0.9$, at most after a budget of 20 fields and never before `min_fields`. If the budget is exhausted with $0.35 < P(f > 0.5) < 0.65$, the call is **inconclusive** rather than forced. The confidence 0.9 and budget 20 were fixed in the label-only simulation (`audit/stopping_rule.py`) before any model existed, and the ± 0.15 band is a round number that was never tuned; none of the three was varied on the test chips. `min_fields` was, and §6.2 reports how it was re-selected without them.

Spread-field order. With n available fields and a budget of k , the rule inspects indices $\text{round}(i \cdot (n-1)/(k-1))$ for $i = 0 \dots k-1$, fields spaced evenly across the acquisition order and read in that order; when $n \leq k$ this is every field, so it differs from "the first k " only on chips larger than the budget.

Metrics. *Field accuracy* is the fraction of fields whose predicted side of 0.5 matches the expert label. *Chip reference* is the majority label of the chip's own fields (for a test chip, the withheld half of its session; a tie counts as good; one test chip, 220606, is tied). *Chip accuracy among called chips* counts every chip that received a *pass* or *fail* — whether the rule stopped at confidence ≥ 0.9 or exhausted the budget outside the inconclusive band; *false-confident rate* is the fraction of called chips whose call was wrong with posterior confidence ≥ 0.9 ; *inconclusive rate* is the fraction that exhausted the budget near 0.5. *Bad-region recall* (sampling simulations) is the fraction of the session's bad fields that the sampled set contains.

Evaluation discipline. Deployed-tool metrics, unless marked otherwise, use a **session-disjoint** test set of 25 sessions (684 fields). These are the withheld half of each held-out session's fields (the controlled-leakage design sets the other half aside; it is never used for training), so a "test chip" is a **session-derived half-chip proxy** — a random half of a date (one or more chips), not a verified physical chip; §6.4 repeats the tool on all 1,377 fields of the same sessions. Where we also report the shipped split's number, it is labelled as such. Leakage A/B (§4.1) and leave-one-cell-line-out (§6.6) use their own splits. Training hyperparameters were fixed before the final evaluation. One decision-rule setting was not: the `min_fields` guard was read off these chips, and §6.2(c) re-selects it without them.

4. Audit

4.1 The published split leaks sessions

Observation. Parsing the archive's central directory shows the split is image-level: sessions are not held out. **Train contains 59 sessions, validation 51, test 57, and train $\cap$ test = 57 of 59 sessions** — images from the same acquisition date appear on both sides.

Controlled measurement. A plain comparison of splits confounds leakage with differences in test composition, so we hold everything else fixed:

- the **test images** are identical in both arms (20 sessions, 494–712 fields depending on the seed);
- the **training-set size is identical** in the two arms (1,440–1,876 fields depending on the seed; the 200 validation fields are identical too and outside both training sets);
- the **model and budget are identical** (MobileNetV3-small [4], ImageNet-initialised, 6 epochs, 224 px, AdamW 3e-4 constant, weight decay 0.01, batch 64, flip augmentation — lighter than the shipped model's recipe, so this is not a re-score of the published split);
- the only difference is whether the training set may use *the other images of the test sessions* (leaky; they make up 27–50% of its training fields, depending on the seed) or must come from disjoint sessions (disjoint). Always predicting the commoner label scores 53.0% on these test sets (mean over seeds).

arm	accuracy	AUC
session-disjoint (mean $\pm$ s.d. over seeds)	67.4% ($\pm$ 6.8 pp)	0.753
same sessions in training (leaky)	75.2% ($\pm$ 3.3 pp)	0.842
inflation (paired, 8 seeds)	+7.9 pp [4.2, 11.5]	+8.9 pp [6.7, 11.0]

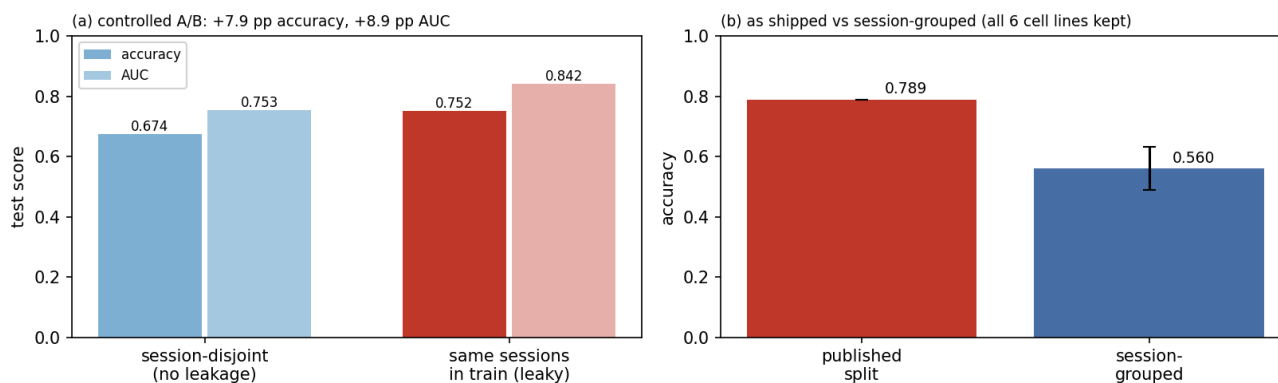


Figure 2. (a) controlled A/B (8 seeds; paired inflation +7.9 pp accuracy, +8.9 pp AUC; positive in every seed — accuracy 4.2–17.8 pp, AUC 6.7–15.0 pp — and the AUC gain grows with the leaked share, $r = 0.74$); (b) shipped vs session-grouped split.

For reference, the shipped split yields **78.9% $\pm$ 0.07 pp / AUC 0.873** (mean $\pm$ s.d. over three seeds; the variation is tiny because all test sessions are seen in training), while a session-grouped split that keeps all six cell lines yields **56.0% $\pm$ 7.2 pp / AUC 0.684** — below the 65.7% of always predicting its commoner label (55.6% on the shipped test set) — a 22.9 pp gap that mixes leakage with the small number of held-out sessions (six). We therefore treat the **controlled A/B as the headline number** and report the shipped/grouped pair only as context.

Why this matters for the benchmark. Any model compared on the shipped split is compared on a test set whose sessions it has already seen. Model *rankings* may or may not survive (we compared one architecture, so we cannot say whether the leak favours some models), but the *absolute* numbers, and any claim of "generalisation to new physical chips", do not.

4.2 Labels are clustered along the acquisition order

Runs test. For each session we take the label sequence in field order and count runs (maximal contiguous blocks of equal label). Under independence the expected run length is $1 / (2 p (1-p))$; with $p \approx 0.56$ that is **2.03** fields. Observed: **6.08** (that pools single-label sessions; within the 45 mixed sessions runs average 5.42 fields versus 2.50 with the same labels in random order). A Wald–Wolfowitz test is significant ($p < 0.05$) in **23 of the 45 mixed sessions**, and 36/45 have negative z (clustered) — against 19.0 and 2.1 expected if each session's labels were in random order (2,000 permutations per session). The most extreme session has $z = -11.9$ (21 runs where 107.3 were expected, 215 fields).

session	fields	good share	runs	expected runs	z	p
230425	215	0.55	21	107.3	-11.93	<1e-6
230419	124	0.07	2	16.0	-10.71	<1e-6
230320	229	0.77	27	82.5	-10.34	<1e-6
230529	216	0.43	48	106.6	-8.18	<1e-6
230317	100	0.73	10	40.4	-7.78	<1e-6
230315	94	0.55	13	47.5	-7.23	<1e-6
230517	90	0.60	12	44.2	-7.11	<1e-6
230314	50	0.44	3	25.6	-6.57	<1e-6
230424	108	0.51	21	55.0	-6.57	<1e-6

Table 1. The nine most clustered sessions, by z (the last two tie; 15 of the 45 mixed sessions have $p < 1e-6$). Expected runs are what independence would give for the same good/bad counts.

This is not a cell-type artifact. A session can contain several cell types imaged in contiguous blocks, which would create clustering for trivial reasons. Restricting the test to *maximal contiguous same-cell-type stretches* (at least 8 fields and both labels present; 72 stretches from 41 sessions), **38 of 72 remain significant** (57/72 clustered).

It is not duplicate frames either. In the six largest sessions, downscaled correlation between consecutive fields is 0.58–0.74 (random pairs in the same sessions: 0.06–0.36; over the six largest and six smallest sessions, mean 0.54 vs 0.28), i.e. consecutive fields are nearby in acquisition order — but counting a new view whenever a field's correlation with the previous one falls below 0.95 gives **1,161 distinct views among 1,183 images (98%)**. The clustering is therefore not redundancy; **failures occupy contiguous stretches of the acquisition order**, not a mapped XY spatial block.

Effective sample size. With $ICC = 0.321$, the design effect $1 + (m-1) \cdot ICC$ is 17.0 with the ANOVA $m \approx 51$ (≈ 181 labels) and 38.2 with Kish's $\tilde{m} \approx 117$, the one for **label prevalence** (≈ 80 ; inverse-variance weighting gives ≈ 167): roughly **80–180 independent labels** for the label mean — not an accuracy or AUC sample size, and not a limit on learning. The mean lag-1 autocorrelation is 0.32 (53% of sessions above 0.3).

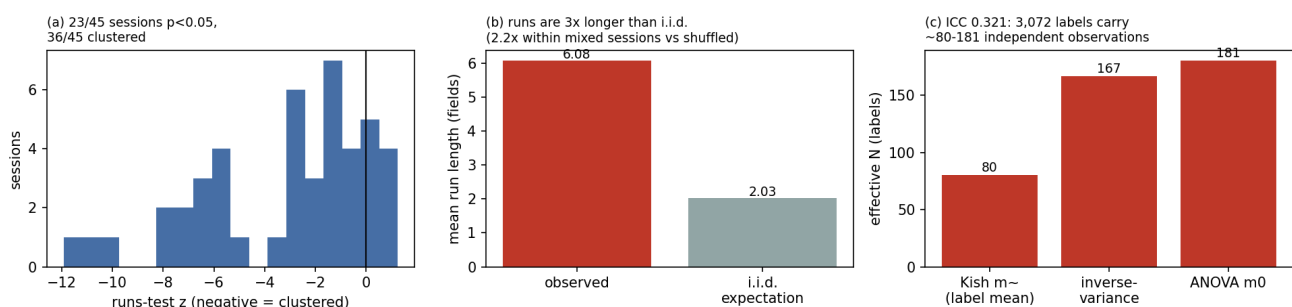


Figure 3. (a) runs-test z per session; (b) observed versus i.i.d. run length, pooled over all sessions; (c) effective sample size under three cluster-size conventions (§3).

Agreement degrades with culture age and is cell-line dependent:

culture age	fields	single-field agreement
0-1_days	852	0.805
2-3_days	798	0.779
4+_days	1199	0.676
4_days	223	0.704

cell line	fields	single-field agreement
A549	775	0.710
CACO	346	0.792
HPMEC	1462	0.702
HSAEC	244	0.840
HUVEC	107	0.991
NHBE	138	0.826

Table 2. Single-field agreement with the session's majority label, by culture age and cell line, counting every field once (74.1% over all 3,072 fields; the 78.8% in the text gives each session equal weight, see below). Late cultures are the hardest, and no cell line is uniformly easy.

Consequence for labels, not just for statistics. If the label were a property of the chip, a single field would predict the session's majority label almost perfectly. It does not: a randomly drawn field agrees with the session majority **78.8%** of the time, three fields 84.1%, five 87.4%, and even ten fields only **91.0%** (up to 50 random draws per session, so each session weighs about equally; counting every field once instead gives 74.1% for one field). Agreement is worse for late cultures (4+ days: 67.6% for a single field) than for day 0–1 (80.5%). The "quality" of a chip is therefore **a property of a stretch of the acquisition order and of the sampling**, not of an image.

4.3 Sampling policy: a comparison that does not survive the model in the loop

If failures are contiguous, how should fields be chosen? We first simulate three policies on the measured **label sequences** at a fixed budget k (300 trials per session):

- **random** — k distinct uniform fields;
- **window** — one contiguous run of k fields (a scan);
- **adaptive** — start at a random field; if it was bad, expand to an unsampled neighbour; if good, jump elsewhere.

budget k	random: acc / recall / err	window: acc / recall / err	adaptive: acc / recall / err	commoner label
2	0.777 / 0.098 / 0.182	0.787 / 0.095 / 0.220	0.772 / 0.103 / 0.203	0.661
4	0.835 / 0.197 / 0.119	0.819 / 0.186 / 0.173	0.803 / 0.214 / 0.154	0.661
8	0.887 / 0.351 / 0.075	0.879 / 0.331 / 0.130	0.828 / 0.391 / 0.121	0.661
12	0.916 / 0.430 / 0.055	0.890 / 0.413 / 0.107	0.830 / 0.475 / 0.109	0.694
20	0.927 / 0.400 / 0.046	0.885 / 0.380 / 0.107	0.781 / 0.482 / 0.138	0.697

Table 3. Label-based policy simulation. $err = |estimated - true\ bad\ fraction|$; last column = calling every chip ($\geq k$ fields) its commoner label; a tied vote (even k) is broken at random. Random looks best for the call from $k = 4$ and adaptive best for localisation — and we do not believe this table.

This is exactly the practice criticised in §6.7, so we repeat the comparison with the deployed model's per-field probabilities on the 25 session-disjoint test chips (`audit/03c_policy_model_in_loop.py`):

budget k	random: acc / recall	window: acc / recall	adaptive: acc / recall	commoner label
4	0.791 / 0.335	0.785 / 0.321	0.772 / 0.323	0.667
8	0.789 / 0.300	0.818 / 0.311	0.776 / 0.296	0.667
12	0.800 / 0.393	0.818 / 0.389	0.840 / 0.397	0.615

Table 4. The same policies with the model in the loop (chips with at least k fields: 24, 15 and 13 at $k = 4, 8, 12$; last column as in Table 3). The ordering changes — adaptive, the label simulation's worst caller at $k \geq 8$, is nominally best at $k=12$ — but nothing survives a multiple-comparison correction. A paired bootstrap over chips (10,000 resamples, `audit/03d_policy_bootstrap.py`) gives 95% intervals at $k=12$ (13 chips) of $[-0.08, -0.00]$ for random – adaptive, $[-0.13, +0.07]$ for window – adaptive and $[-0.13, +0.08]$ for random – window. The first only just excludes zero and rests on the 4 of 13 chips where the two policies differ at all. With a Bonferroni correction for the six differences tested (three pairs $\times k = 8, 12$) no interval excludes zero — that one's upper bound is exactly 0 (Figure 4c).

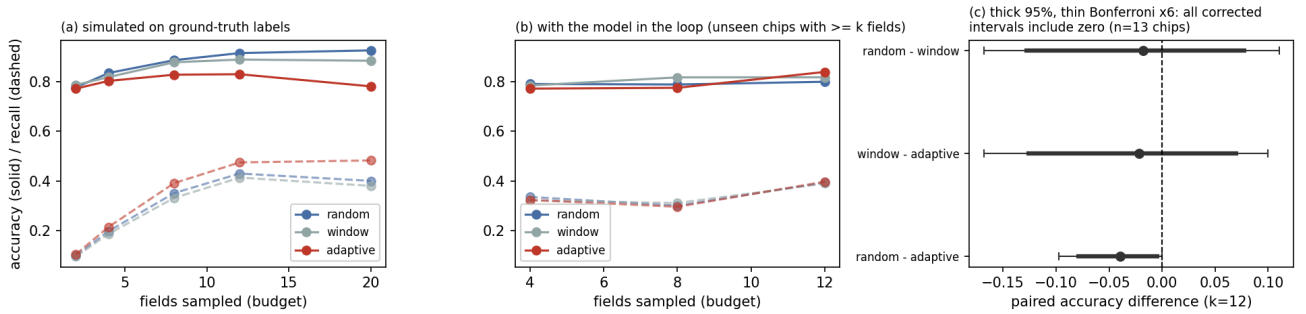


Figure 4. Policy comparison. (a) simulated on ground-truth labels; (b) with the model in the loop; (c) paired bootstrap differences at $k=12$ — thick bars 95%, thin bars Bonferroni-corrected for six comparisons; every corrected interval includes zero, so no ranking is claimed.

We therefore report no policy ranking. With only 13–15 chips large enough for these budgets the comparison is underpowered, and the label-based ranking is an artefact of scoring policies against the very labels they were designed to sample.

What *is* established, and what the tool uses, is a single reproducible failure of the obvious default. Reading the **first** k fields in acquisition order with the sequential rule of §6.1 called a 100%-bad chip (session 230405) *pass* with 0.94 confidence: the per-field probabilities in the first third of that session are low (mean 0.255 over the first five fields, versus 0.935 over the last twenty). Sampling fields spread across the chip fixes that case (the same chip returns *fail*, $P = 0.91$); on the 11 chips with at least 20 fields the accuracy difference (0.727 vs 0.636) is ± 1 chip and is not significant either. The justification for spread sampling is the observed failure, not a measured ranking.

4.4 The same defect in a second organoid benchmark

To test whether the leakage we measure is a property of one dataset or of the field, we audited a second public organoid imaging benchmark: the **OCT organoid segmentation-and-tracking dataset** (zenodo.15783866, CC-BY-4.0; Branciforti et al., *Diagnostics* 2024 [5,6]). Its file names encode the acquisition group — `w<well>_d<day>_s<slice>.png` in train/val and `d<day>_p<plate>_w<well>_s<slice>.png` in test (all test images are plate 1; train/val names carry no plate). If train/val come from the same plate — the names give no

other one — the same (well, day) means the same organoids imaged in the same session. Reading the archive's central directory over HTTP range requests (no 4.9 GB download; `audit/06_oct_leakage.py`):

- train: 5,584 images in 8 (well, day) groups; val: 1,396 images in 2 groups; test: 6,980 images in 10 groups (each image is stored three times — original, manual mask, automatic mask; we count originals)
- **train $\cap$ test = 4 groups** — well 2 at days 5, 7, 11 and 13; val $\cap$ test = 1 group
- **40.0% of test images (2,792 of 6,980) belong to a (well, day) group that also appears in training**

For a *tracking* benchmark this is more severe than for classification: under that reading the same organoid instances, imaged in the same session at the same timepoint, appear on both sides of the split. The file names cannot rule out a second plate with the same well numbering; the paper's own split description would settle it. We report the structural overlap rather than a re-trained inflation number, because re-training their segmentation/tracking pipeline is out of scope here.

Two datasets, two independent instances of the same defect — group-level splits that leak — which is the empirical basis for the first rule of the corrected protocol (§5).

4.5 A hypothesis we tested and rejected: re-image vs discard

The dataset's own definition of `bad` mixes **technical artifacts** (bubbles, defocus, deformed walls) with **biological problems** (morphology/density off expectation) [1]. If these could be separated, the actionable output would be "re-image" (cheap, the chip survives) versus "discard" (expensive, re-imaging is useless). Two observations encouraged the attempt: isolated bad runs are blurrier and darker than good fields (single-field runs: mean absolute gradient 0.089 vs 0.126; dark fraction 0.137 vs 0.094 — bubbles and defocus), whereas long bad runs are sharper (0.150) and higher in contrast (s.d. 0.243 vs 0.199; `audit/04b_run_image_stats.py`); and visually, 4/4 sampled isolated-bad fields showed bubbles or defocus.

We constructed a target — a run of `bad` fields "recovers" if at least three of the next five fields are `good`; runs followed by fewer than five fields, including the 30 of 250 that reach the end of the session, count as not recovering (base rate 62.4%) — and tried to predict it from features available at decision time (run length, whether the run reaches the end of the session, focus, darkness, artifact scores), with **session-grouped cross-validation**.

feature set	CV AUC
run length only	0.476
artifact signals only	0.381
run length + artifacts	0.539
all features	0.613
all features except <i>reaches the end</i>	0.539
permutation null	mean 0.479, p95 0.564
simple rule ($\text{length} \leq 2 \Rightarrow \text{recover}$)	accuracy 0.560 vs base rate 0.624

The decision rule is not supported. The target is confounded by position (because failures are contiguous, the fields after a run are usually good, so "recovery" largely means "the run did not reach the end"; without that feature the model scores 0.539), and — more fundamentally — **no re-imaging was ever performed in this dataset, so the causal question cannot be observed**. We therefore ship `pass/fail/inconclusive` and document the two-mode observation as an *open problem*, not a feature.

4.6 Aggregation rules are a sensitivity/specificity dial

If several fields are read, their scores must be combined. We compare three rules on per-field probabilities from a session-disjoint model (a lighter 224 px, 6-epoch model than the shipped one, with its field threshold chosen on validation rather than the tool's 0.5), with the chip reference defined as before:

budget k	mean: acc / sens / spec	max: acc / sens / spec	top-2: acc / sens / spec	commoner label
1	0.704 / 0.456 / 0.832	0.704 / 0.456 / 0.832	0.704 / 0.456 / 0.832	0.660
3	0.691 / 0.475 / 0.803	0.699 / 0.691 / 0.704	0.709 / 0.610 / 0.760	0.660
5	0.678 / 0.493 / 0.767	0.626 / 0.686 / 0.598	0.627 / 0.574 / 0.653	0.683
8	0.742 / 0.645 / 0.787	0.627 / 0.878 / 0.507	0.660 / 0.787 / 0.598	0.686
12	0.742 / 0.851 / 0.710	0.570 / 1.000 / 0.354	0.631 / 0.997 / 0.452	0.671

Table 5. Aggregation rules (2 seeds, each its own split of 25 held-out sessions; 60 random k-field draws per chip, shared by all rules; chips with < k fields excluded; last column as in Table 3). Field thresholds, chosen on validation by balanced accuracy: 0.30 and 0.25.

`mean` maximises chip accuracy and keeps specificity high; `max` maximises sensitivity (it calls almost everything bad: specificity collapses to 0.35 at k=12) and `top-2` sits between. **The aggregation rule moves the operating point more than the per-field model does** — at k=12, accuracy ranges from 0.570 to 0.742 and sensitivity from 0.851 to 1.000 depending only on the rule. From k = 5, `max` and `top-2` fall below the commoner-label floor (last column); `mean` clears it only at $k \geq 8$. A QC deployment must therefore state its aggregation rule; "accuracy" without it is not a specification.

We also tested a rule that exploits the measured autocorrelation — smoothing the per-field scores along the acquisition order before aggregating. It did **not** help: on the shipped predictions a 3- or 5-field moving average changes chip accuracy by at most 0.007 at any budget from 4 to 12 fields (k=6: 0.792 → 0.795 with a 3-field and 0.793 with a 5-field window; `audit/smoothing_test.py`). Reported as a negative result.

4.7 Metadata does not explain the labels

A logistic model on day, log seeding density, flow rate and cell line reaches only 0.562 in-sample accuracy on the 2,213 fields with complete metadata, against 0.507 for always predicting the majority label. Cell lines do differ on average (Caco-2 68.5% bad vs NHBE 23.9%), but every cell line has sessions with a 0% bad rate and sessions with 100%. The label is dominated by **session-level** variation, consistent with §4.2.

5. Corrected evaluation protocol

From the audit, three rules follow for anyone training or benchmarking on this dataset:

1. **Split by session, not by image.** No session may contribute to two splits. Report the number of held-out sessions alongside the metric — with 59 sessions, a 20% held-out split is ~12 sessions and the confidence interval on chip-level accuracy is correspondingly wide.
2. **Evaluate at the chip level.** Report (i) per-field accuracy/AUC *and* (ii) chip-level accuracy with the number of fields used, because per-field numbers do not answer the lab's question and are far less certain than their count suggests under within-session correlation (label-prevalence effective $N \approx 80$ –180, not 3,072; not an accuracy/AUC N).
3. **State the sampling policy.** Which fields were used, and how many. Policies differ in principle (§4.3) and we could not rank them with this benchmark's 59 sessions; a number is only interpretable together with the policy that produced it.

As a worked example, the scripts listed in §10 write the result files that pin named headline numbers and the numeric result tables enumerated by `check_tables.py`.

6. The tool

6.1 Design

`inference.py` takes the fields of one chip (filenames in acquisition order) and returns:

- **per-field $P(\text{bad})$** from a MobileNetV3-small (ImageNet-initialised, 384 px, 25 epochs, cosine LR, label smoothing, random-resized-crop/flip augmentation), trained on a **session-disjoint** split (1,495 train / 200 val / 684 test fields from 25 unseen sessions);
- a **chip call** — `pass`, `fail`, or `inconclusive` — from a Beta(1,1) posterior on the bad fraction, stopping when $P(\text{chip bad}) > 0.9$ or < 0.1 ;
- **how many fields were needed**, with fields read from an **evenly spaced grid** (every field in order on chips of ≤ 20 fields);
- a **QC map** (field index vs $P(\text{bad})$) and a `chip_report.json`.

Model and training. MobileNetV3-small [4], ImageNet-initialised, classifier head replaced with a 2-way linear layer. Input 384×384 (the source images are $2,056 \times 1,542$). AdamW (lr $3e-4$, weight decay 0.02), cosine schedule to lr/30, label smoothing 0.05, batch 32, 25 epochs (≈ 14 s/epoch on one RTX 3090, ≈ 6 min total), RandomResizedCrop(0.7–1.0) + horizontal/vertical flips. Label smoothing and the modest capacity are deliberate: the labels are four-rater majorities, and 3,072 fields with a label-prevalence effective N of ~ 80 –180 do not support a large model. The checkpoint is shipped in `model/`.

6.2 Two bugs found during deployment, and how their fixes were chosen

Deploying the tool on real chips exposed two failure modes that a label-only simulation had hidden.

(a) Reading the first k fields is wrong. On a 100%-bad chip, walking the first fields in acquisition order stopped after six fields (five `good` predictions, one `bad`) and returned `pass` with 0.94 confidence. Because failures are contiguous (§4.2), the beginning of a session can lie inside a good region. Sampling fields spread across the chip fixed this specific case (the same chip now returns `fail` at 0.91).

(b) The Beta posterior is overconfident on few fields. With three good fields and no bad one, $P(\text{chip bad}) = 0.0625$ — "94% confident" — which is indefensible when the per-field model is imperfect. Requiring a minimum of eight fields before a stop is allowed cut the false-confident rate from 25% to 13% — but it cannot act on the 10 chips with fewer than eight fields (Table 7).

(c) The value 8 was read off the test chips, and it does not survive re-selection. Both fixes above were found on the 25 test chips, and the minimum of eight was chosen from a sweep on them (Figure 5). To see how much of the test gain is selection, we re-selected it without the test chips (`audit/09_inner_cv_minfields.py`): five-fold session-grouped cross-validation over the 34 non-test sessions with the shipped training recipe, each session cut into two half-chips like the test chips (68 OOF half-session proxies; models retrained per fold, so clustered, not 68 new physical chips), and a criterion fixed before looking — highest chip accuracy with every chip called, ties to fewer fields.

on 68 non-test chips (out-of-fold)	min 1	min 8	min 12	all fields
chip accuracy, every chip called	0.779	0.765	0.765	0.765
confident-but-wrong chips	11	10	8	—
fields per chip	5.3	8.1	9.9	24.9

On these 68 chips, calling every chip pass scores 0.662.

The criterion selects **min 1**, whose test result is **0.680** with every chip called (0.708 on the 24 it calls, 6.8 fields). The guard's accuracy gain on the test chips (0.680 → 0.800) is therefore not reproduced on independent physical chips and should be read as optimistic; on those OOF half-session proxies the guard only trims confident errors (11 → 8 of 68 at min 12) for one chip of accuracy. What *does* replicate is the efficiency claim: at every minimum the rule matches reading every field (0.765–0.779 versus 0.765) with 5.3–9.9 fields instead of 24.9, and in the deferring mode it is 0.806 on the 62 of 68 chips it calls at min 8 (0.800 on 65 at min 1) — close to the 0.826 measured on the test chips. We keep min 8 in the tool for the a-priori reason in (b) — three agreeing fields cannot justify 94% (a reason that holds only on chips of eight or more fields) — and report the test-chip numbers at 8 below with this caveat. The same re-selection picks the plain cap of Table 6 as **20**, not 12 (0.765 on the 68 chips, 0.760 on the test chips with 12.6 fields), and reading the first fields instead of spread ones is no better there (0.750 vs 0.765 at min 8): the spread order is kept for the failure in (a), not for a measured gain.

6.3 Measured performance (25 unseen chips, 684 fields)

metric	value
per-field accuracy / AUC	0.734 / 0.791 (always predicting good: 0.525)
chip accuracy, all fields, mean	0.80 (calling every chip <i>pass</i> : 0.64, 16 of 25)
chip accuracy among called chips (spread fields, min 8 — tuned on these chips, §6.2(c))	0.826 (95% Wilson CI 0.63–0.93, n=23)
false-confident calls (conf ≥ 0.9 and wrong, of 23 calls)	13% (3/23)
inconclusive (budget exhausted near P = 0.5)	8%
mean fields used	9.5 (a plain cap of 12 spread fields: 9.0 fields, 0.80)

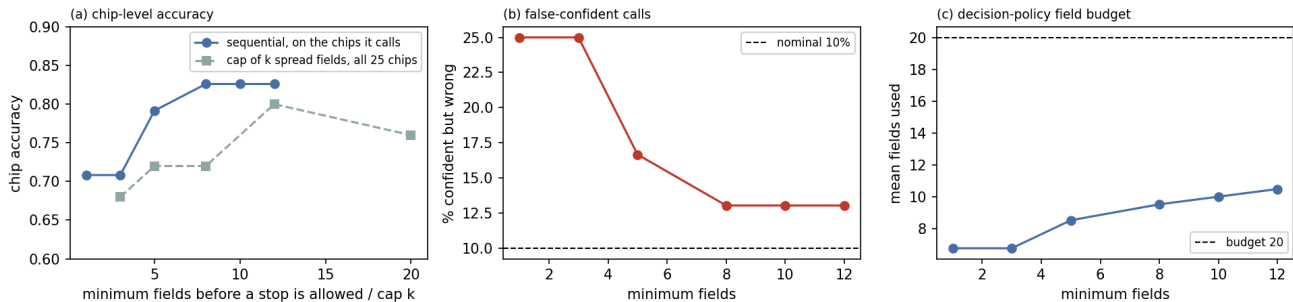


Figure 5. Tool behaviour versus the minimum-fields guard, swept on the 25 test chips (the sweep the value 8 was read from; §6.2(c) re-selects it without them). In (a) the sequential curve is accuracy on the chips it calls (23–24 of 25); the grey curve is a cap of k spread fields on all 25.

More capacity or resolution did not help (one seed each). We trained a backbone with 2.8× the parameters (MobileNetV3-large, same 384 px, same split and schedule; batch 8 instead of 32 to fit memory). Test field accuracy moved from 0.734 to 0.746 and balanced accuracy from 0.733 to 0.747, while AUC *fell* from 0.791 to 0.788 — i.e. a larger model buys nothing here (audit/perfield_model.py --arch large). Higher resolution does not either: the small backbone at 512 px (batch 12) gives accuracy 0.734 and AUC 0.797 (--size 512). Together with the session-level shift in §6.5 (almost every unseen session sits outside the training feature distribution) this suggests the labels and chip-to-chip appearance limit this benchmark more than model capacity does (not a proven ceiling: other architectures and training recipes were not tried). We therefore ship the smaller model, which is cheaper to run; with the same stopping rule the larger model uses the same 9.5 fields and is not more accurate on the chips it calls (0.792 vs 0.826).

6.4 Field efficiency: the same accuracy with 2.9× fewer fields

The naive way to use a per-field model is to read every field of the chip and average. On the 25 test chips that costs **27.4 fields per chip for 0.800 accuracy**. The shipped rule — fields spread across the chip, Beta posterior, stop at confidence 0.9, minimum 8 fields — reaches **0.800 on all 25 chips using 9.5 fields (2.9× fewer)**, and when it is allowed to defer the chips it cannot resolve it reaches **0.826 on the 23 it calls, deferring 8%** (`audit/07_efficiency.py`):

policy	fields per chip	chips called	accuracy
all fields (mean of every field)	27.4	25/25	0.800
fixed k = 12, spread (chips with ≥ 12 fields only)	12.0	13/25	0.846
cap of 12 spread fields (all fields if fewer)	9.0	25/25	0.800
sequential, force (min 8)	9.5	25/25	0.800
sequential, abstain (min 8)	9.5	23/25	0.826

Table 6. Field efficiency. The fixed-k row is evaluated only on the 13 chips that have at least 12 fields, so it is not comparable with the others; the cap and sequential rows use every chip. A plain cap of 12 fields is as frugal as the sequential rule, but the right cap is only known after the fact (a cap of 8 gives 0.720, a cap of 20 gives 0.760; selected without the test chips the cap would be 20, §6.2(c)). The min-8 rows share the same caveat: without the test chips the minimum would be 1, which scores 0.680 here; the rule's contribution is the per-chip confidence, the pass / fail / inconclusive call and the option to defer, not a further cut in fields.

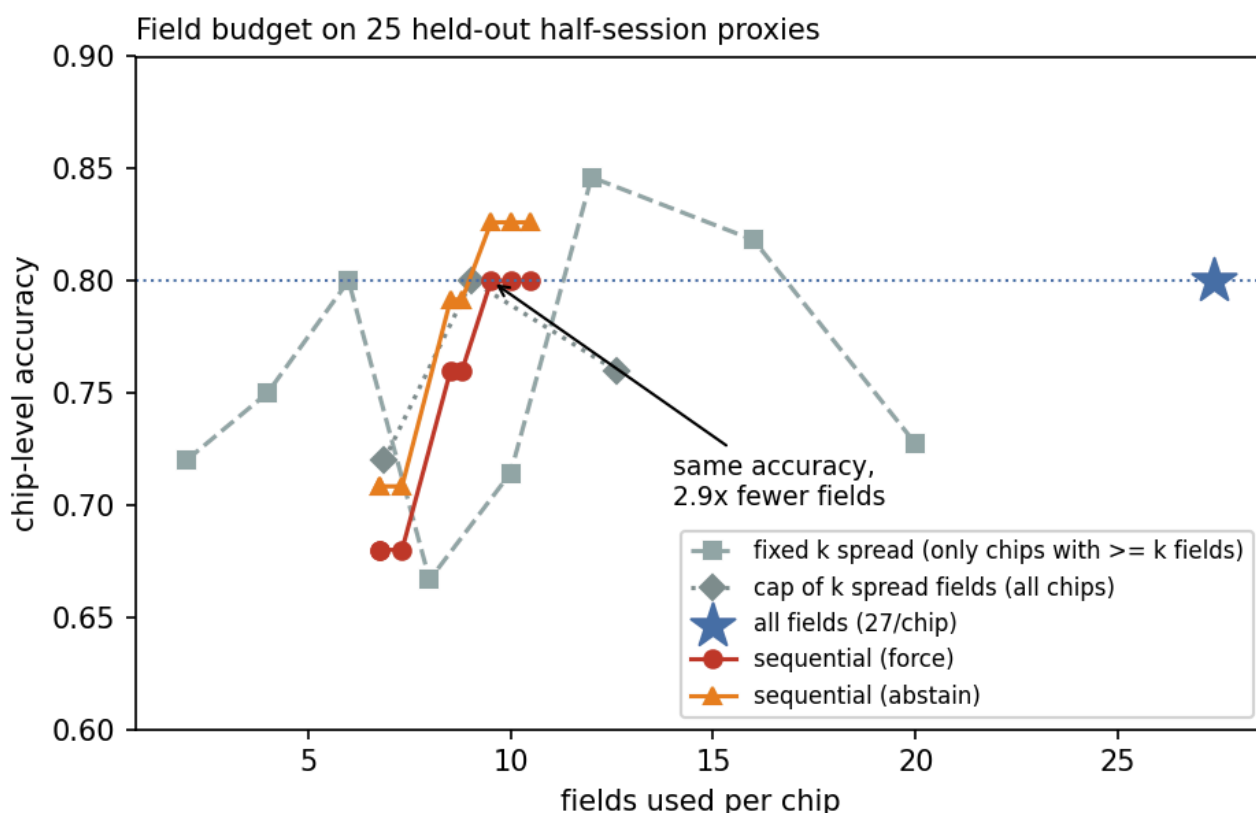


Figure 6. Accuracy against the label budget per chip (fields the policy consumes on a scored map, not imaging cost).

For a lab imaging plates of chips this is the practical *decision-budget* number: the policy consumes roughly a third of the fields of the read-everything workflow on already-scored maps (not measured microscope, operator, or inference time), and 8% of chips are flagged "needs a human" instead of being guessed.

Full sessions. On all fields of the same 25 held-out sessions (1,377 fields, 55.1 per chip, reference = full-session majority; `audit/08_full_sessions.py`) the saving grows and the accuracy drops: reading every field gives 0.76, the shipped rule gives **0.76 with 9.0 fields (6.1× fewer)**, but it defers no chip and 6 of its 25 calls are confident and wrong (min 10 or 12: 0.80, 11–12.6 fields). The half-session numbers above are therefore the favourable end.

The tool ships this as a **plate mode** (`inference.py --plate <folder>`; one subfolder per chip), which returns a ranked triage table. On the 25 test proxies it consumes **238 of 684 scored fields (65% of the decision budget unused)** — offline QC-map consumption, not microscope time — and returns 6 *fail*, 2 *inconclusive* and 17 *pass*, worst first.

6.5 Where the tool fails

The model's own confidence does not flag chip-level failures: all four wrong calls in Table 7 were made at confidence 0.89 or higher, three of them above the 0.9 stopping threshold. The clearest case is session 230405 (69 fields, **100%** labelled bad): the model is right on 82.6% of its fields individually, yet the per-field probabilities in the first third of the session are low (mean 0.255 over the first five fields versus 0.935 over the last twenty), so an unguarded rule stopped early with a confident *pass*.

chip	fields	bad share	field acc		chip	fields	bad share	field acc
220706	5	100%	0.200		220506	7	29%	0.857
220718	4	100%	0.750		220715	7	29%	0.429
220721	3	100%	1.000		230123	13	23%	0.692
230405	69	100%	0.826		220515	6	17%	0.833
230403	6	83%	0.167		230109	53	13%	0.792
230314	25	64%	0.720		230214	24	12%	0.667
230512	44	61%	0.795		220502	4	0%	1.000
230529	108	59%	0.769		220620	6	0%	1.000
230523	55	55%	0.582		220716	4	0%	0.750
220606	10	50%	0.200		230119	35	0%	0.971
230425	107	49%	0.673		230316	22	0%	1.000
230315	47	47%	0.638		230321	8	0%	1.000
230215	12	42%	0.583					

Table 7. The 25 unseen test chips, sorted by the true share of bad fields (left column, then right). The deployed rule (spread fields, min 8, conf 0.9) makes four wrong calls — 220706 (100% bad, 5 fields) and 230403 (83% bad, 6 fields) called pass with confidence 0.89 and 0.99, 230314 (64% bad) pass at 0.91, 230425 (49% bad) fail at 0.93 — and returns inconclusive on 220606 and 230529. Chips with fewer than 8 fields never reach the min-8 guard; they account for 2 of the 4 wrong calls (2 of 10 such chips, versus 2 of 15 larger ones).

Out-of-distribution detection failed. We tested two standard detectors against the training fields: a Mahalanobis distance on four image statistics (mean, standard deviation, mean absolute gradient as a focus proxy, dark fraction) and a Mahalanobis distance on the model's 576-dimensional pooled backbone feature, each alarming above the training 99th percentile (`audit/10_ood_check.py`). The image-statistics detector flags none of the 25 unseen chips. The feature-space detector flags 21 of 25 when a chip is scored by its median field distance (almost every unseen session looks shifted) and 4 of 25 when scored by its mean feature. **None of the three flags session 230405**, the worst failure. We therefore do not present an OOD alarm as a safety feature; the distance is exposed as a diagnostic only.

What this means in practice. The tool is usable as a *screening* aid with a human in the loop, not as an autonomous gate. Its measured error rate — 4 of 23 calls wrong (17%), 3 of them at ≥ 0.9 confidence (13%) — is the number to quote in any deployment decision.

6.6 Generalisation to a cell line the model has never seen

The session-disjoint protocol of §6.3 keeps all six cell lines in training. To measure a stricter setting we hold out one cell line entirely — test = all of its fields, train = the other five with any session containing the held-out line removed — and retrain the deployed architecture and input size for 25 epochs (batch 16 and a cosine floor of $1e-5$ instead of 32 and $lr/30$, no validation split, one seed, inherited train sizes). Loader: Resize+flips, **no** crop, different `drop_last` than the shipped recipe — the AUC gap is not an isolated cell-line effect (`audit/05_lolo_cellline.py`):

held-out cell line	test fields (sessions)	accuracy (majority label)	balanced acc	AUC
A549	775 (24)	0.712 (0.693)	0.661	0.678
CACO	346 (17)	0.512 (0.685)	0.611	0.666
HPMEC	1462 (29)	0.589 (0.546)	0.605	0.659
HUVEC	107 (4)	0.879 (0.860)	0.874	0.930
NHBE	138 (6)	0.601 (0.761)	0.499	0.630
HSAEC	244 (21)	0.730 (0.668)	0.723	0.753
mean of the six	3,072 (101)	0.670	0.662	0.719
in-distribution (all lines seen, §6.3)	684 (25)	0.734	0.733	0.791

Table 8. Leave-one-cell-line-out. The spread between cell lines (AUC 0.630–0.930) is larger than the effect of either modelling change we tested — $2.8\times$ the parameters or 512 px inputs moved AUC by at most 0.006 (§6.3). Brackets: always predicting the line's commoner label — Caco-2 and NHBE fall below it.

In this separate evaluation, a held-out cell line costs about seven AUC points on average and up to 22 accuracy points (Caco-2, against the pooled 0.734; against Caco-2's own 89 fields in the session-disjoint test set, $0.742 \rightarrow 0.512$), and the per-line variation dominates. Measured line by line in the same way, the AUC drop averages 16 points over the five lines whose test fields contain both labels — but those baselines rest on 13 to 344 fields each, so the pooled 7 points is the steadier estimate. Not a matched-recipe ablation; we keep the session-disjoint protocol as the headline because it matches how such a tool would be used (a new chip of a known cell line), but any deployment to an unseen line needs re-calibration on that line.

6.7 Label-only simulation overstates deployed performance

A common practice — which we followed first — is to simulate a decision policy on **ground-truth labels**. Our first one (random order, no minimum, all 59 chips) gave 92.8% with 6.6 fields; rule and chips differ from deployment, so we also run the *deployed* rule on the *same* 25 chips with label calls and with model calls (`audit/label_vs_model_same_rule.py`, Figure 7). Without a minimum, labels give $21/24 = 0.875$ (one confident-but-wrong) and the model $17/24 = 0.708$ (six); with the 8-field minimum, 0.875 (8.5 fields) and $19/23 = 0.826$ (9.5 fields, three) — a gap the minimum closes on the chips it was tuned on (§6.2(c)).

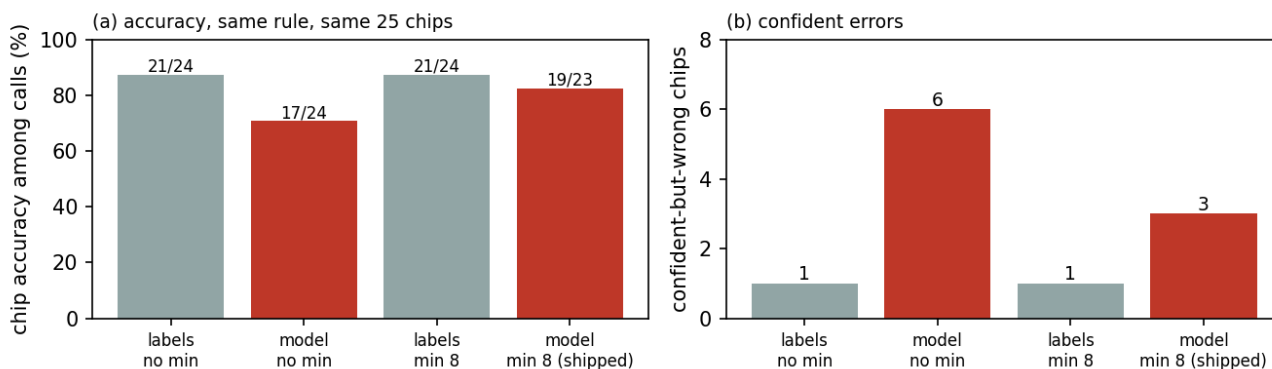


Figure 7. The same rule on the same 25 chips: ground-truth field labels versus model calls.

Recommendation for the field: evaluate QC decision policies **with the model in the loop**. A label-only simulation measures the policy, not the system.

6.8 Interactive demo

A browser-side build of the tool (ONNX Runtime Web, no server) is available at <https://taewoong23-ooc-chip-qc-demo.static.hf.space/index.html>. Bundled-example buttons use Python-reference probabilities. The corrected hosted upload path (demo/browser/ source) accepts a restricted still-PNG format and displays exploratory field scores only, suppressing operational chip calls and confidence. Bundled-reference replay and live scores are explicitly separate. This is not prospective deployment validation or a validated chip-quality instrument. ONNX vs PyTorch on identical tensors is max $|\Delta P|$ 8.5e-06 (audit/12_onnx_parity.py); that does not cover the live path. The model is sensitive to the resize method — bicubic, box or non-antialiased bilinear instead of the training resize flip 11, 19 and 35 of 200 test field calls (audit/13_preprocessing_sensitivity.py). The bundled examples are stored at 1,024 × 768, on which the borderline chip is called *pass* ($P = 0.33$) where its original fields give *inconclusive*.

7. Application value

Laboratory automation. The daily QC pass is the most repetitive imaging task in a chip lab: every chip is looked at, most are fine, and the decision is made by eye. The tool returns a ranked plate triage — on 25 test proxies it flags 6 failures and defers 2, while consuming 238 of 684 scored fields (65% of the decision budget unused; not microscope time) — so a human starts with the chips that most need attention (passes are not guaranteed: 3 of 23 calls are confident and wrong).

Data assetisation and standardisation. The audit's most transferable result is that a chip imaging dataset's *information content* is not its file count: 3,072 fields carry a **label-prevalence** effective N of roughly **80–180 independent labels** (not an accuracy/AUC N), and a published split that leaks acquisition groups can inflate accuracy by **7.9 pp**. Organisations building chip data assets, or training models on them, can use the same measurements to judge what their data is worth and when a reported number can be believed. The corrected protocol (session-level splits, chip-level metrics, explicit sampling policy) does not depend on this dataset; we have checked the split defect in two benchmarks, not more.

Drug evaluation and toxicology. QC gates every downstream readout: a chip-level layer with an *inconclusive* outcome can screen failing chips out before dose-response curves (a human confirms, given 3 of 23 confident wrong calls) and makes each exclusion auditable.

Toward chip digital twins. The tool's output — a pass/fail state with a posterior confidence, a map of where the culture degrades, and its cost (9.5 fields) — is a first culture-level state estimate for a twin.

What we do not claim. No wet-lab validation, no biological or clinical validity, one dataset for the tool (§9). The contribution is a *measurement and decision layer* whose error rate is stated.

8. Discussion

For benchmark authors. Three checks are cheap and the published benchmark did not report them: a session-level (or patient- or chip-level) split, a design-effect estimate for the label, and the agreement of a single sample with its group's consensus. They change what an accuracy means: here the label-prevalence effective sample size is 3–6% of the nominal one, and the shipped split tests on seen sessions.

For laboratories. With clustered failures, *which* fields are read matters: the same rule calls a long chip right or wrong depending on whether it reads the start or the whole acquisition order. We could not rank sampling policies with 59 sessions (§4.3); the evenly spaced grid avoids the one failure we observed. Its cost is 9.5 fields; its error rate (4 of 23 calls wrong, 3 confidently) is what matters.

For method developers. Per-field errors interact with early stopping (Figure 7: 0.875 on labels, 0.708 with the model, same rule and chips, no minimum). Policy numbers from label simulations should be read with this in mind; model-in-the-loop evaluation should be the default.

Open problem. The dataset's own definition of bad mixes technical artifacts with biological failure, which suggests the actionable output should be "re-image" versus "discard". We tried and failed to validate that mapping (§4.5): the causal question cannot be observed without actual re-imaging. A dataset that images each chip twice — before and after a re-imaging attempt — would settle it, and would be a small, valuable addition to this benchmark.

9. Limitations

1. **4 of 23 calls wrong on unseen session-derived half-chip proxies, 3 of them at ≥ 0.9 confidence (13%).** The confidence is a posterior under an independence assumption, not a calibrated probability (mean 0.93 over the 23 calls; 0.826 correct). A research prototype, not a validated operational instrument: no chip should be discarded on its output alone.
2. **No reliable out-of-distribution detector.** Image-statistics and feature-space distances both missed the worst case (§6.5); the tool exposes the distance as a diagnostic only.
3. **The re-image/discard recommendation is not validated** (§4.5) and is not part of the tool's output.
4. **One dataset, 25 session-derived half-chip proxies.** Chip-level metrics carry wide confidence intervals; we report them as point estimates with the split size. Each test unit is the withheld half of a held-out date (one microscope run over one or more chips); on the full sessions the rule is 0.76 accurate with 6 of 25 confident-but-wrong (§6.4). No wet-lab validation was performed, and we make no biological or clinical claim — the claim is about *measurement validity*. The tool accepts single-chip folders; these numbers are on the proxies.
5. **Sampling policies could not be ranked.** The label-based comparison (§4.3, Table 3) is underpowered with the model in the loop (13–15 chips; no paired difference survives a correction for six comparisons). We report the policy we use and why, not a ranking.
6. **Labels come from a four-rater majority**, and inter-rater agreement is not public, so we cannot say how close any accuracy here is to the ceiling the labels allow; some of the measured error may be label noise rather than model error.
7. **Deployment is unvalidated, including on cell lines seen in training.** The separate leave-one-cell-line-out evaluation gives mean AUC 0.719 versus 0.791 (§6.6); its different training recipe and data composition prevent attributing that gap solely to an unseen cell line. The deployed tool was evaluated on this dataset's session-derived proxies, not prospectively validated on physical chips.
8. **The acquisition order is used as a proxy for reading order, not as proof of a spatial block.** Consecutive fields are highly correlated ($r = 0.58\text{--}0.74$ vs $0.06\text{--}0.36$ for random pairs in the six largest

sessions), which supports the order proxy; XY stage geometry is not public.

9. **The minimum-fields guard was tuned on the test chips.** The shipped value 8 comes from a sweep on the 25 test chips; re-selected on the 34 other sessions it would be 1 (test accuracy 0.680 instead of 0.800 with every chip called). The headline 0.826 / 13% are therefore optimistic by an unknown amount; the field-budget saving is observed on these proxies, not a non-inferiority result or a guarantee on new physical chips (§6.2(c)). Nor is its margin over calling every chip *pass* established: test McNemar is 5 vs 1 discordances, $p = 0.22$. Validation has 12 vs 5 discordant halves from 34 session clusters; those counts are descriptive, with no iid p reported.

10. Reproducibility

```
pip install -r requirements.txt # then download the dataset and datasheet as in README
§1

python3 audit/leakage_experiment.py # published split: 57/59 sessions shared;
shipped vs grouped (§4.1)
python3 audit/leakage_controlled.py # controlled A/B, +7.9 pp / +8.9 pp over 8 seeds
 (§4.1)
python3 audit/01b_leakage_ci.py # paired 95% CIs over the 8 seeds (§4.1)
python3 audit/structure_probe.py # runs test, Table 1; metadata model (§4.2,
 §4.7)
python3 audit/02c_celltype_control.py # runs test within same-cell-type stretches
 (§4.2)
python3 audit/02d_redundancy.py # consecutive vs random field correlation,
distinct views (§4.2)
python3 audit/block_structure.py # ICC 0.321, design effect 17 (ANOVA m0) (§4.2)
python3 audit/15_effective_n.py # effective N 80 (Kish) – 181 (ANOVA m0) (§4.2)
python3 audit/label_sufficiency.py # single-field agreement, Table 2 (§4.2)
python3 audit/adaptive_sampling.py # policy comparison on labels, Table 3 (§4.3)
python3 audit/03c_policy_model_in_loop.py # ... with the model in the loop, Table 4
 (§4.3)
python3 audit/03d_policy_bootstrap.py # paired bootstrap CIs over chips, Figure 4c
 (§4.3)
python3 audit/06_oct_leakage.py # second benchmark, 40% of test images leak
 (§4.4)
python3 audit/recovery_test.py # re-image vs discard, negative result (§4.5)
python3 audit/04b_run_image_stats.py # focus/darkness of isolated vs long bad runs
 (§4.5)
python3 audit/aggregation_experiment.py # mean / max / top-2 aggregation, Table 5 (§4.6)
python3 audit/smoothing_test.py # smoothing along acquisition order, negative
 (§4.6)
python3 audit/perfield_model.py # trains the shipped per-field model (§6.1; --
arch large, --size 512)
python3 evaluate.py # deployed-tool numbers, Figure 5 (§6.3)
python3 audit/07_efficiency.py # field efficiency, Table 6 (§6.4)
python3 audit/08_full_sessions.py # the tool on full sessions (§6.4)
python3 audit/09_inner_cv_minfields.py # min-fields guard re-selected without the test
chips (§6.2(c))
python3 audit/00_ood_reference.py && python3 audit/10_ood_check.py # OOD reference,
detectors (§6.5)
python3 audit/11_per_chip_calls.py # per-chip calls, Table 7; the first-k failure
 (§4.3, §6.2(a))
python3 audit/05_lolo_cellline.py # leave-one-cell-line-out, Table 8 (§6.6)
python3 audit/12_onnx_parity.py && python3 audit/13_preprocessing_sensitivity.py #
```

```
demo checks (§6.8)
python3 audit/14_vs_all_pass.py          # rule vs all-pass, paired (§9)
python3 audit/label_vs_model_same_rule.py # labels vs model, same rule and chips (§6.7)
python3 audit/stopping_rule.py           # the original label-only stopping simulation (§3, §6.7)
python3 figures.py                       # every figure in this report
python3 demo/app.py                      # interactive demo
```

Named headline anchors map to `results/*.json`; the enumerated numeric result tables are checked against their JSON values and expected row schemas (`check_claims.py` / `claims.json`, `check_numbers.py`, `check_tables.py`). Every matched anchor is checked; missing required anchors fail, and `--mutations` tests planted errors. This covers the ledger's named claims and the checker's enumerated numeric tables, not every document, table or prose token.

External models, libraries and tools. ImageNet-pretrained MobileNetV3 weights from torchvision (BSD-3-Clause). Tested with PyTorch 2.13, torchvision 0.28, NumPy 2.2, SciPy 1.15, scikit-learn 1.7, Pillow 12.3, Matplotlib 3.10, Gradio 6.20, ONNX Runtime 1.23 (Python) and ONNX Runtime Web 1.20 (browser), on one RTX 3090; the tool runs on CPU. The demo video's narration is synthesised with edge-tts (Microsoft neural voice en-US-AriaNeural). An AI coding assistant (Claude) was used for code, analysis scripts and drafting; named headline numbers are pinned to result keys and every report table is checked cell by cell (`check_numbers.py`, `check_tables.py`).

11. Related work

Leakage in benchmarks. Split leakage is a known class of problem: Ramos et al. audit it across seven visual datasets [7]; whole-slide-image benchmarks leak at patient and institution level, with a measurable accuracy gap [8]; OCT classification was inflated by improper splitting [9]; tooling exists for near-duplicate detection and grouped splitting [10,11]. Ours is a new instance, with the magnitude measured in an OoC QC benchmark by a controlled A/B.

Clustered labels and effective sample size. Treating images as correlated clusters is standard in imaging study design [12]; we apply it to field-level QC **label prevalence** (≈ 80 – 180 independent labels; not an accuracy/AUC N).

Microscopy QC. Automated QC for high-throughput microscopy has its own benchmarks and methods (e.g. AutoQC-Bench, 2025 [13]); classical QC metrics for microscopy images are long established [14]. Our contribution is complementary: not a new QC metric, but a measurement audit and a sampling/decision policy for chip-level QC.

Organoid/OoC imaging benchmarks. Several organoid imaging datasets exist (detection, segmentation, tracking), including multi-rater annotation [15]. None, to our knowledge, ship a chip-level QC protocol; the field-dependence we measure is a property of this label definition.

12. Conclusion

A public OoC QC benchmark reports per-image accuracy on a split that leaks sessions ($+7.9$ pp in a controlled A/B, 8 seeds, lighter recipe), with labels clustered along the acquisition order (ICC 0.321; label-prevalence effective $N \approx 80$ – 180 of 3,072) and field-dependent (one field agrees with the session majority 78.8% of the time). A session-grouped, chip-level protocol yields a tool that matches reading every field with 9.5 decision fields and says "inconclusive" when it cannot decide (82.6% on the chips it calls) — but its 8-field minimum was tuned on

the test chips (0.680 with every chip called without them, §6.2(c)) and its margin over calling every chip *pass* is not significant (§9). The observed gain is a smaller decision-field budget on these proxies, not a non-inferiority or new-chip guarantee. Label-only simulation overstates the rule (0.875 vs 0.708 with the model); neither a re-image/discard split nor a reliable OOD detector was supported.

References

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